

Worksheet 1 — The Math of the Atom

High School · Grades 9–12 · Physics & Algebra II / Pre-Calc · ~45 min

Use the constants and formulas in the reference box to solve each problem. Show your work. These problems mirror the calculations real nuclear engineers, health physicists, and reactor operators perform every day.

Name: _____

Date: _____

Class: _____

Constant / Formula	Value / Expression
Speed of light, c	3.00×10^8 m/s
1 atomic mass unit (u)	1.6605×10^{-27} kg
1 MeV	1.602×10^{-13} J
Avogadro's number	6.022×10^{23} /mol
Mass–energy equivalence	$E = mc^2$
Radioactive decay	$N(t) = N_0 \cdot (1/2)^{(t/T^{1/2})}$
Decay constant	$\lambda = \ln(2) / T^{1/2}$
Reactor multiplication	$k_{\text{eff}} = \text{neutrons produced} / (\text{neutrons absorbed} + \text{leaked})$

1. Mass–Energy Equivalence

A single fission of U-235 converts about 0.215 u of mass into energy. Calculate the energy released (a) in joules and (b) in MeV. Show all unit conversions.

2. Scaling Up

A 1,000-MW(e) reactor operates at ~33% thermal efficiency, so it generates about 3,000 MW of thermal power. Using your answer from Problem 1, estimate the number of U-235 fissions per second required to sustain that thermal output.

3. Fuel Burnup

If each fission consumes one U-235 atom (mass ≈ 235 u), how many kilograms of U-235 are fissioned per day in the reactor from Problem 2? Compare this to the $\sim 9,000$ metric tons of coal a comparable coal plant burns per day.

4. Half-Life Decay

Iodine-131 (a fission product used in medical imaging) has a half-life of 8.02 days. If a hospital receives a 200 mCi (millicurie) source, how much activity remains after 30 days? Express your answer in mCi.

5. Decay Constant

Compute the decay constant λ for Cesium-137 ($T_{1/2} = 30.17$ years) in units of per second. Why is this isotope a major concern in long-term spent-fuel storage?

6. Reactor Criticality

A reactor has $k_{\text{eff}} = 1.002$. (a) Is the reactor subcritical, critical, or supercritical? (b) If each neutron generation lasts ~ 0.1 s (delayed-neutron-dominated), by what factor does the neutron population grow in

60 seconds?

7. Energy Density Comparison

Using your Problem 3 result, calculate the specific energy of U-235 fission in MJ per kilogram and compare to gasoline (~46 MJ/kg). Express the ratio.

8. Dose Estimation (Real-World Application)

Background radiation in the US averages ~3.1 mSv/year. A single chest CT scan is ~7 mSv. A cross-country flight is ~0.04 mSv. Living next to a nuclear plant adds <0.01 mSv/year. Rank these from largest to smallest and discuss which dose source most people underestimate.

Stretch Challenge — College Preview

The four-factor formula approximates k_{∞} in a thermal reactor: $k_{\infty} = \eta \cdot f \cdot p \cdot \epsilon$, where η is the reproduction factor, f the thermal utilization, p the resonance escape probability, and ϵ the fast fission factor. Research each term and write a one-paragraph explanation of how a reactor designer would increase k_{∞} while preserving negative temperature feedback.

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Reactor multiplication	$k_{\text{eff}} = \text{neutrons produced} / (\text{neutrons absorbed} + \text{leaked})$

1. Mass–Energy Equivalence

A single fission of U-235 converts about 0.215 u of mass into energy. Calculate the energy released (a) in joules and (b) in MeV. Show all unit conversions.

Answer: (a) $m = 0.215 \times 1.6605 \times 10^{-27} = 3.570 \times 10^{-28}$ kg. $E = mc^2 = 3.570 \times 10^{-28} \times (3.00 \times 10^8)^2 = 3.21 \times 10^{-11}$ J. (b) $E = 3.21 \times 10^{-11} / 1.602 \times 10^{-13} \approx \mathbf{200 \text{ MeV}}$ per fission.

2. Scaling Up

A 1,000-MW(e) reactor operates at ~33% thermal efficiency, so it generates about 3,000 MW of thermal power. Using your answer from Problem 1, estimate the number of U-235 fissions per second required to sustain that thermal output.

Answer: $P = 3.0 \times 10^9$ J/s. Fissions/s = $3.0 \times 10^9 / 3.21 \times 10^{-11} \approx \mathbf{9.3 \times 10^{19}}$ fissions/s.

3. Fuel Burnup

If each fission consumes one U-235 atom (mass ≈ 235 u), how many kilograms of U-235 are fissioned per day in the reactor from Problem 2? Compare this to the ~9,000 metric tons of coal a comparable coal plant burns per day.

Answer: Atoms/day = $9.3 \times 10^{19} \times 86,400 \approx 8.04 \times 10^{24}$. Mass = $8.04 \times 10^{24} \times 235 \times 1.6605 \times 10^{-27} \approx \mathbf{3.14 \text{ kg}}$ U-235/day — roughly **2.9 million times less mass** than coal.

4. Half-Life Decay

Iodine-131 (a fission product used in medical imaging) has a half-life of 8.02 days. If a hospital receives a 200 mCi (millicurie) source, how much activity remains after 30 days? Express your answer in mCi.

Answer: $N/N_0 = (1/2)^{(30/8.02)} = (1/2)^{3.741} \approx 0.0747$. $A = 200 \times 0.0747 \approx \mathbf{14.9 \text{ mCi}}$.

5. Decay Constant

Compute the decay constant λ for Cesium-137 ($T_{1/2} = 30.17$ years) in units of per second. Why is this isotope a major concern in long-term spent-fuel storage?

Answer: $\lambda = \ln(2)/(30.17 \times 3.156 \times 10^7 \text{ s}) \approx \mathbf{7.28 \times 10^{-10} \text{ /s}}$. Cs-137 emits energetic gamma rays and persists for centuries (10 half-lives ≈ 300 yr to reach 0.1%).

6. Reactor Criticality

A reactor has $k_{\text{eff}} = 1.002$. (a) Is the reactor subcritical, critical, or supercritical? (b) If each neutron generation lasts ~ 0.1 s (delayed-neutron-dominated), by what factor does the neutron population grow in 60 seconds?

Answer: (a) **Supercritical**. (b) Generations = 600. Growth = $1.002^{600} \approx e^{(600 \cdot \ln 1.002)} \approx e^{1.199} \approx \mathbf{3.32 \times}$. Operators trim k_{eff} back toward 1.000.

7. Energy Density Comparison

Using your Problem 3 result, calculate the specific energy of U-235 fission in MJ per kilogram and compare to gasoline (~ 46 MJ/kg). Express the ratio.

Answer: E per kg = $200 \text{ MeV} \times 1.602 \times 10^{-13} \times (6.022 \times 10^{23} / 0.235) \approx \mathbf{8.2 \times 10^{10} \text{ J/kg} \approx 82 \text{ million MJ/kg}}$. Ratio to gasoline $\approx \mathbf{1.8 \text{ million } \times}$.

8. Dose Estimation (Real-World Application)

Background radiation in the US averages ~ 3.1 mSv/year. A single chest CT scan is ~ 7 mSv. A cross-country flight is ~ 0.04 mSv. Living next to a nuclear plant adds < 0.01 mSv/year. Rank these from largest to smallest and discuss which dose source most people underestimate.

Answer: CT (7) > background (3.1) > flight (0.04) > nuclear-plant neighbor (< 0.01). Most people underestimate medical and background sources, which dwarf plant emissions.

Stretch Challenge — College Preview

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Sample response: Designers tune fuel enrichment (raises η), reduce parasitic absorbers (raises f), optimize moderator-to-fuel ratio (raises p), and select fuel/moderator combinations that maintain a negative Doppler broadening coefficient so heating naturally reduces reactivity.

Worksheet 2 — Reactor Systems Engineering

High School · Grades 9–12 · Engineering, Thermodynamics, Systems Thinking · ~50 min

Engineers design nuclear plants as systems of systems. In this worksheet you will analyze thermodynamic efficiency, defense-in-depth safety, and the trade-offs between competing reactor designs — the same trade-spaces practicing nuclear, mechanical, and electrical engineers navigate.

Name: _____

Date: _____

Class: _____

Part A · Thermodynamic Efficiency

The Carnot efficiency sets the upper bound for any heat engine: $\eta_{\text{Carnot}} = 1 - T_{\text{cold}} / T_{\text{hot}}$, with temperatures in Kelvin.

Reactor type	T_hot (°C)	T_cold (°C)	η_{Carnot}	Real η
PWR (Gen II)	315	30	_____	~33%
BWR (Gen II)	285	30	_____	~33%
High-Temp Gas Reactor	750	30	_____	~45%
Molten Salt Reactor	700	30	_____	~44%
Sodium Fast Reactor	550	30	_____	~40%

A1. Compute η_{Carnot} for each row. **A2.** Why is the real efficiency always far below Carnot? Identify at least three real-world loss mechanisms. **A3.** Which design is the best candidate for industrial process heat (hydrogen production, desalination, chemical synthesis) and why?

Part B · Defense in Depth

Nuclear safety relies on independent, redundant barriers. Match each safety feature with the failure it is designed to mitigate.

Safety feature	Failure it mitigates
1. Fuel pellet ceramic matrix (UO ₂)	A. Loss of off-site power (station blackout)
2. Zircaloy fuel cladding	B. Sudden reactivity insertion / power excursion
3. Reactor pressure vessel (steel, 8 in thick)	C. Release of fission products to environment
4. Containment building (reinforced concrete + steel liner)	D. Loss-of-coolant accident (LOCA)
5. Negative temperature coefficient	E. First barrier: traps fission products in ceramic
6. Emergency core cooling system (ECCS)	F. Mechanical impact, aircraft strike, external hazards
7. Passive natural-circulation cooling (Gen III+)	G. Second barrier: contains gaseous fission products

Your matches: 1__ 2__ 3__ 4__ 5__ 6__ 7__

Part C · Reactor Design Trade Study

You are a junior engineer on a team selecting a reactor for a remote Alaskan community of 5,000 people with a 20 MW peak demand. Score the three candidates on a 1–5 scale across each criterion, justify with one sentence, then recommend one design.

Criterion	Large PWR (1100 MW)	SMR (BWRX-300, 300 MW)	Microreactor (10 MW)
Capital fit to load			
Site footprint			
Refueling logistics			
Construction time			
Operator staffing			
Total (out of 25)			

C1. Which design did you recommend and why?

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Reactor type	T_hot (°C)	T_cold (°C)	η_{Carnot}	Real η
PWR (Gen II)	315	30	48.5%	~33%
BWR (Gen II)	285	30	46.3%	~33%
High-Temp Gas Reactor	750	30	70.4%	~45%
Molten Salt Reactor	700	30	68.8%	~44%
Sodium Fast Reactor	550	30	63.2%	~40%

A1. Compute η_{Carnot} for each row. **A2.** Why is the real efficiency always far below Carnot? Identify at least three real-world loss mechanisms. **A3.** Which design is the best candidate for industrial process heat (hydrogen production, desalination, chemical synthesis) and why?

A2 sample: turbine blade inefficiencies, condenser ΔT losses, pump and auxiliary parasitic loads, steam-cycle entropy generation, generator losses. **A3:** The HTGR at 750 °C delivers high-grade heat usable for sulfur-iodine or high-temperature electrolysis hydrogen production.

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Answer key: 1→E, 2→G, 3→D, 4→C, 5→B, 6→D, 7→A.

Part C · Reactor Design Trade Study

You are a junior engineer on a team selecting a reactor for a remote Alaskan community of 5,000 people with a 20 MW peak demand. Score the three candidates on a 1–5 scale across each criterion, justify with one sentence, then recommend one design.

Criterion	Large PWR (1100 MW)	SMR (BWRX-300, 300 MW)	Microreactor (10 MW)
Capital fit to load	1 (oversized 50×)	2 (still oversized)	5 (matched)
Site footprint	1 (large)	3 (modest)	5 (small)
Refueling logistics	2 (18-mo cycle, large outage)	3 (factory fuel)	5 (5-10 yr cassette)
Construction time	1 (5–10 yr)	3 (3–5 yr)	5 (<2 yr, factory)
Operator staffing	1 (~600 staff)	3 (~75 staff)	5 (~5 staff, remote-monitored)
Total (out of 25)	6	14	25 — recommended

C1. Which design did you recommend and why?

Worksheet 3 — Your STEM Pathway in Nuclear

High School · Grades 9–12 · Career, College & Self-Assessment · ~40 min

Nuclear energy is one of the most interdisciplinary fields in STEM — it needs physicists, materials scientists, software engineers, electricians, machinists, policy analysts, communicators, and more. This worksheet helps you see where your own strengths and curiosities could lead.

Name: _____ Date: _____ Class: _____

Part A · Map the Discipline to the Job

STEM area you enjoy	Nuclear-industry role	Typical degree path	Median US salary (2020)
Calculus + physics	Reactor / nuclear engineer	BS/MS Nuclear or Mech. Eng.	\$125k
Chemistry + materials	Fuel cycle / materials scientist	BS/PhD Mat. Sci. or Chem.	\$110k
Computer science	Reactor simulation / digital twin development	BS CS or Computational Eng.	\$130k
Electrical / electronics	I&C engineer, grid integration	BS Electrical Eng.	\$115k
Biology + safety	Health physicist, radiation protection	BS/MS Health Physics	\$105k
Hands-on / mechanical	Reactor operator, NDT technician	AS + NRC license	\$100k
Writing / policy	Regulatory analyst, nonproliferation	BA + MA Policy / Law	\$95k
Business / finance	Project finance, utility planning	BS/MBA Finance or Econ.	\$120k

Salary figures are rough US estimates compiled from BLS, ANS career data, and industry surveys; actual figures vary by region, employer, and experience.

Part B · Self-Inventory

1. Which two STEM areas above appeal to you most right now? Why?

2. Which one is the biggest *stretch* for you — and would you be willing to grow into it?

3. Of the eight roles above, which one would you most want to job-shadow for a day? What three questions would you ask that professional?

4. Pick a current nuclear company (e.g., Westinghouse, NuScale, X-energy, Helion, Radiant, Terrestrial Energy, Constellation Energy). Visit their careers page and list three open positions plus the

qualifications each requires.

5. Identify one high school class you can take next semester that moves you toward your top choice (AP Physics, AP Chemistry, AP Calculus, AP Computer Science, Engineering Design, etc.).

Part C · Build a Four-Year Plan

Sketch a high school course sequence that prepares you for a nuclear-adjacent STEM major. Add one extracurricular or summer experience per year (robotics, science fair, internship, DOE-NE student programs, ANS Student Conference, etc.).

Year	Math	Science	Tech / Eng	Extracurricular / Summer
9th				
10th				
11th				
12th				

Part D · Resources to Explore

- American Nuclear Society (ANS) student programs — ans.org/students
- U.S. DOE Office of Nuclear Energy — energy.gov/ne
- Nuclear Regulatory Commission Student Programs — nrc.gov
- Nuclear Energy Institute (NEI) Careers — nei.org/careers
- Idaho National Laboratory (INL) internships — inl.gov/careers
- MIT OpenCourseWare 22.01 — Intro to Nuclear Engineering (free)
- edX / Coursera — search 'nuclear energy' for free university courses

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Teacher note

Part B and C are open-ended self-assessment — there is no single correct answer. Suggested rubric: depth of reflection (40%), realistic alignment between interests and chosen role (30%), specificity of four-year plan (30%).